How-To Library

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Testing Security Rules with the Security Rule Set Simulator

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Abstract

This article describes how to create masking and blocking security rules in Dynamic Data Masking and test the rules with the Security Rule Set Simulator.

Supported Versions

- Dynamic Data Masking 9.6.1 or later

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Overview

You can use the Security Rule Set Simulator to test rules that you create in the Management Console.

Your organization has an Oracle production database that contains sensitive information about employees, such as names and salaries. You want to block access to the sensitive information for most of the database users, but you want the database administrator to view masked data.

In this example, you complete the following tasks:

1. Create a connection to the database in the Management Console.
2. Add the Dynamic Data Masking service for Oracle in the Management Console and create a connection rule that directs requests to the security rule set.
3. Create the security rule set.
4. Create a masking rule that masks data that the database administrator accesses.
5. Create a blocking rule that blocks access to the database for users that are not the database administrator.
6. Test the security rule set as a database administrator to verify that the administrator views masked data.
7. Test the security rule set as a database user to verify that the user views an error message instead of the sensitive data.

Step 1. Create the Database Connection

Create a connection to the database that you want to restrict access to.

1. In the Management Console, select the root domain node and click **Tree** > **Add Database**.

   The **Add Database** window appears.

2. Select the Oracle database type and configure the following database parameters:

DDM Database Name

Enter the logical name for the database. The Dynamic Data Masking database name appears in the Management Console tree. In this example, the database name is HRDatabase.

Instance Name

Enter the instance name for the target database.

Listener Address

Enter the server host name or TCP/IP address for the target database.

Listener Port

Enter the TCP/IP listener port for the target database.

Service Name

Enter the service name for the target database. Dynamic Data Masking determines the target database based on the service name or SID in the client connection request.

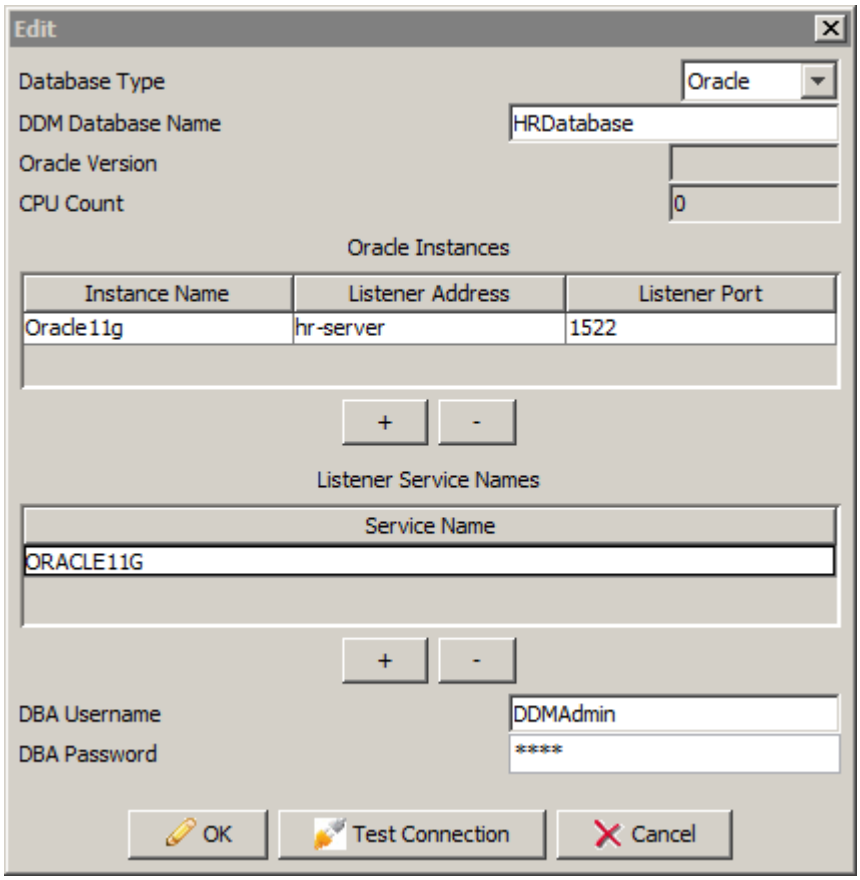
DBA Username

Enter the user name for the database administrator account to log in to the Oracle database. You must use an administrator account so you can test the masking rule that you create for the database administrator. In this example, the database administrator user name is DDMAdmin.

DBA Password

Enter the password for database administrator.

The following image shows the database connection parameters:



3. Note the **DDM Database Name** and **DBA Username** values. You use the database connection parameters when you run the Security Rule Set Simulator.
4. Click **Test Connection** to validate the connection to the database.
5. Click **OK**.

For more information about database connections, see the *Dynamic Data Masking 9.6.1 Administrator Guide*.

Step 2. Create the Connection Rule

Add the Dynamic Data Masking Oracle service and create a connection rule that directs incoming requests to the Oracle database.

1. In the Management Console, select the Dynamic Data Masking Server node and click **Tree** > **Add DDM Services**.

   The **Add DDM Services** window appears.
2. In the **Add DDM Services** window, select the **DDM for Oracle** service and click **OK**.

   The **DDM for Oracle** service appears in the Management Console tree.
3. Select the **DDM for Oracle** service node and click **Tree** > **Connection Rules**.

   The **Rule Editor** window appears.
4. In the Rule Editor, click **Action** > **Append Rule**.

   The **Append Rule** window appears.

5. Enter the following connection rule parameters:

   Rule Name

   Enter the name of the rule. The rule name appears in the rule tree. In this example, the connection rule name is DirectToRuleSet.

   Matcher

   Select the All Incoming Connections matcher. The All Incoming Connections Matcher applies the connection rule action to all incoming database requests.

   Action

   Select the Use Rule Set rule action. The Use Rule Set action directs the requests to the security rule set that you define in the **Rule Set Name** field.

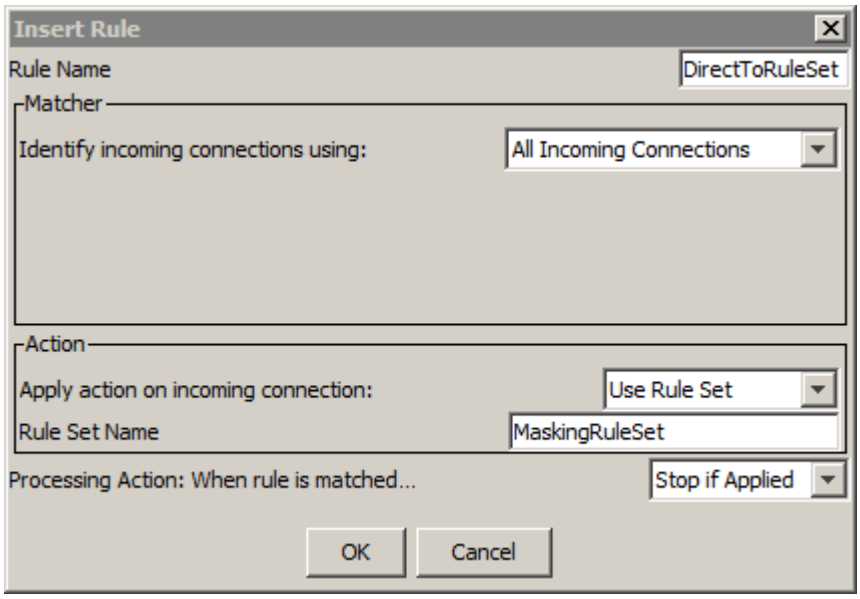
   Rule Set Name

   Enter the name of the security rule set that you want to create to restrict access to the database. In this example, the rule set name is MaskingRuleSet.

   Processing Action

   Select the Stop if Applied processing action. The Stop if Applied processing action instructs the Rule Engine not to continue to the next rule in the tree.

   The following image shows the connection rule parameters:



6. Note the **Rule Set Name** value that you entered for the connection rule. You use the value to name the security rule set.
7. Click **OK** to close the window.
8. Click **File** > **Update Rules** to save the connection rule and click **File** > **Exit** to close the Rule Editor.

Step 3. Create the Security Rule Set

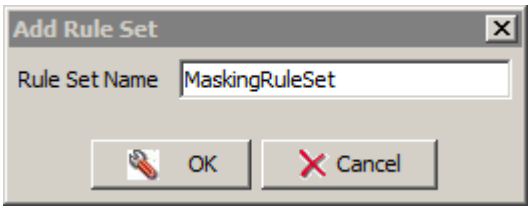
Create the security rule set to hold the blocking and masking rules.

1. In the Management Console, select the root domain node and click **Tree** > **Add Rule Set**.

   The **Add Rule Set** window appears.

2. In the **Rule Set Name** field, enter the name of the rule set. Enter the same name that you entered in the connection rule **Rule Set Name** field. In this example, the rule set name is MaskingRuleSet.

   The following image shows the **Add Rule Set** window:



3. Click **OK**.

   The rule set appears in the Management Console tree.

Step 4. Create the Masking Rule

Create a masking rule that masks data that the database administrator accesses.

1. In the Management Console tree, select the security rule set that you created and click **Tree** > **Security Rule Set**.

   The **Rule Editor** window appears.

2. In the Rule Editor, click **Action** > **Append Rule**.

   The **Append Rule** window appears.

3. Configure the following parameters for the masking rule:

   Rule Name

   Enter the name of the masking rule. In this example, the rule name is MaskingRule.

   Description

   Enter an optional description of the rule.

   Matcher

   Select the Symbol matcher. In the **Symbol Name** field, enter AUTH_USERNAME. The Rule Engine can use the AUTH_USERNAME symbol to identify the incoming request based on the user that makes the request. In the **Text** field, enter user name of the administrator. In this example, the administrator user name is DDMAdmin.

   Action

   Select the Mask action. When you select the Mask action, the Rule Engine applies the masking function that you define. Click the plus sign (+) to add a masking function. In this example, the masking function is the default, `substr(\(col),1,2)`. The default masking function leaves the first two characters in every column unmasked and removes every other character.

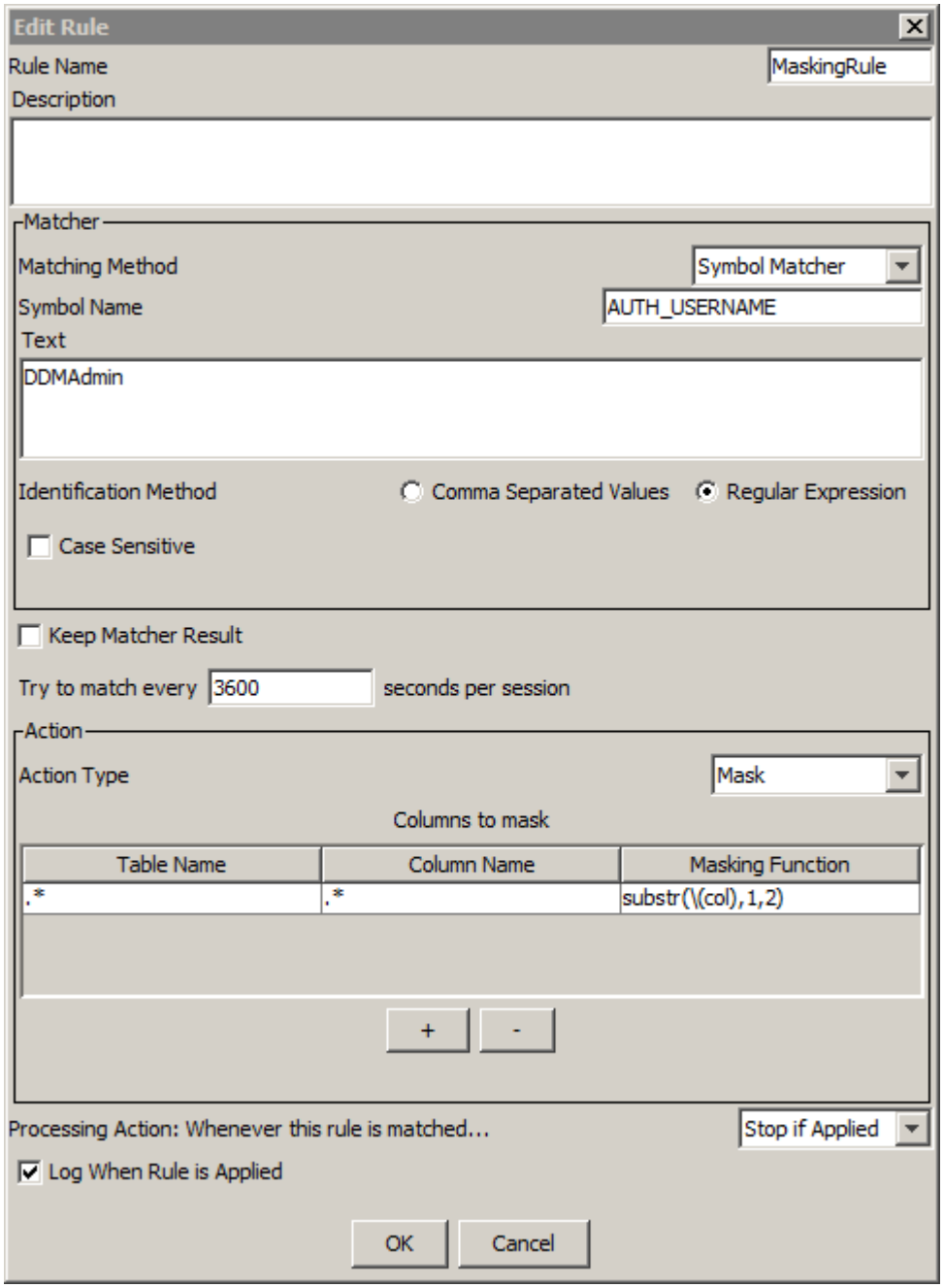
   Processing Action

   Select the Stop if Applied processing action. The Rule Engine does not continue to the next rule in the security rule set if it matches the masking rule and applies the masking function.

   Log When Rule is Applied

   Select the **Log When Rule is Applied** parameter to include information in the log files about when the Rule Engine applies the masking rule.

The following image shows the masking rule parameters:



4. Click **OK** to close the Rule Editor.
5. Click **File** > **Update Rules** to save the rule.

Step 5. Create the Blocking Rule

Create a blocking rule to block access to the database for users that are not the database administrator.

1. In the Rule Editor, select the masking rule that you created and click **Action** > **Insert Rule**.

   The **Insert Rule** window appears.

2. Configure the following parameters for the blocking rule:

   Rule Name

   Enter the name of the blocking rule. In this example, the rule name is BlockingRule.

   Description

   Enter an optional description of the rule.

   Matcher

   Select the Any matcher. The Any matcher matches all incoming connections. Because the masking rule uses the Stop if Applied processing action, the only requests that go through the blocking rule are requests that are made by someone other than the database administrator.

   Action

   Select the Block Statement rule action. In the **Error Message** field, enter a message that the user sees when the Rule Engine blocks the request. In this example, the error message is "DDM blocks access to this data."

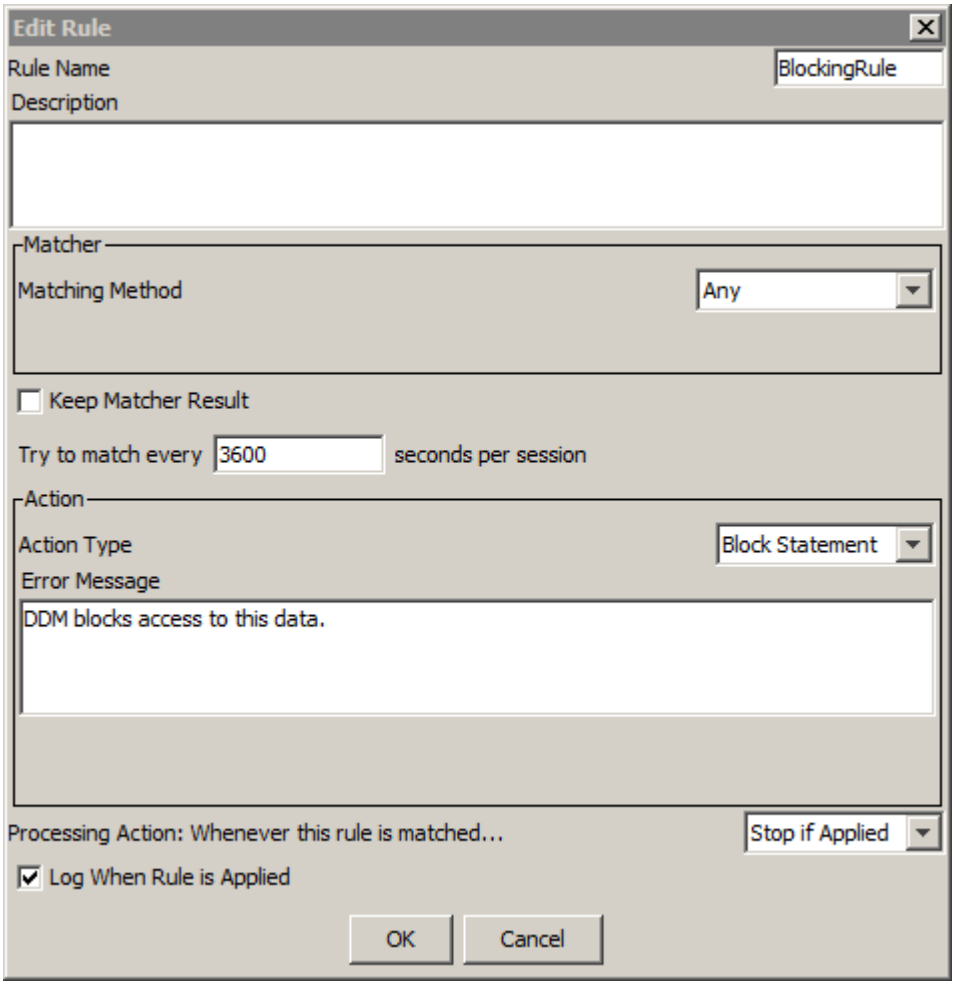
   Processing Action

   Select the Stop if Applied processing action.

   Log When Rule is Applied

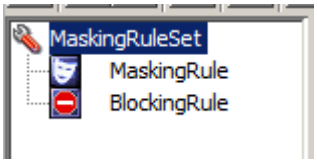
   Select the **Log When Rule is Applied** parameter to include information in the log files about when the Rule Engine applies the blocking rule.

   The following image shows the blocking rule parameters:



3. Click **OK** to close the Rule Editor.
4. Click **File** > **Update Rules** to save the rule.

   The following image shows the security rule set tree:



Step 6. Simulate the Rule Set

Simulate the rule set to test whether you created the rules correctly.

You use the Security Rule Set Simulator to view simulated log output and verify that the Rule Engine alters the SQL request and masks or blocks data the way you need. In the Security Rule Set Simulator, you define the database and provide database administrator login information so that Dynamic Data Masking can access the database and provide

accurate log output. You enter a sample SQL request and the Rule Engine runs through the rule set as if the database user made the request from a third party client.

This example runs the simulator as a database administrator and a database user to verify that the Rule Engine correctly recognizes the users and applies the appropriate security rule for each user.

Simulating the Rules as an Administrator

Run the Security Rule Set Simulator as an administrator to view how the Rule Engine applies the masking rule to the request.

1. In the Rule Editor, select the rule set node in the rule tree and click **Action** > **Security Rule Set Simulator**.

   The Security Rule Set Simulator opens.

   Note: This example uses the simulator to test the rule set. To test an individual rule, select the rule in the rule tree and click **Action** > **Security Rule Set Simulator**.

2. Configure the following parameters in the Security Rule Set Simulator:

   DDM Database Name

   Enter the name of the Dynamic Data Masking database that you defined when you created the connection to the database. In this example, the database name is HRDatabase.

   DBA Username

   Enter the name of the database user that you used to connect to the database. In this example, the user name is DDMAdmin.

   DBA Password

   Enter the password for the database user.

   Symbol Definition

   In the AUTH_USERNAME row of the **Symbol Definition** field, enter the name of the database administrator that you want to receive masked results. In this example, the database administrator is DDMAdmin.

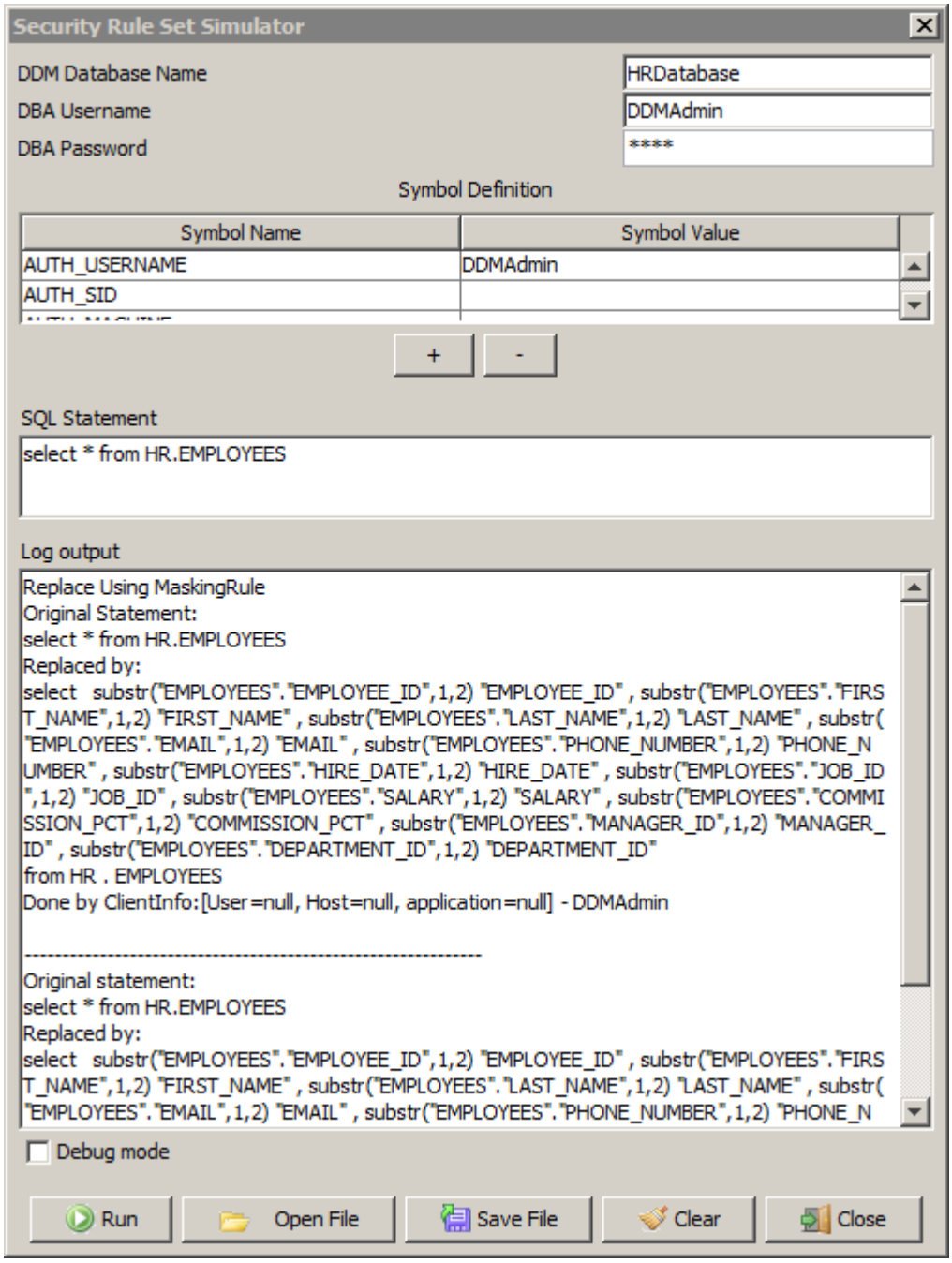
   SQL Statement

   Enter the SQL request that you want to use to test the rule. In this example, the SQL request is `select * from HR.EMPLOYEES`.

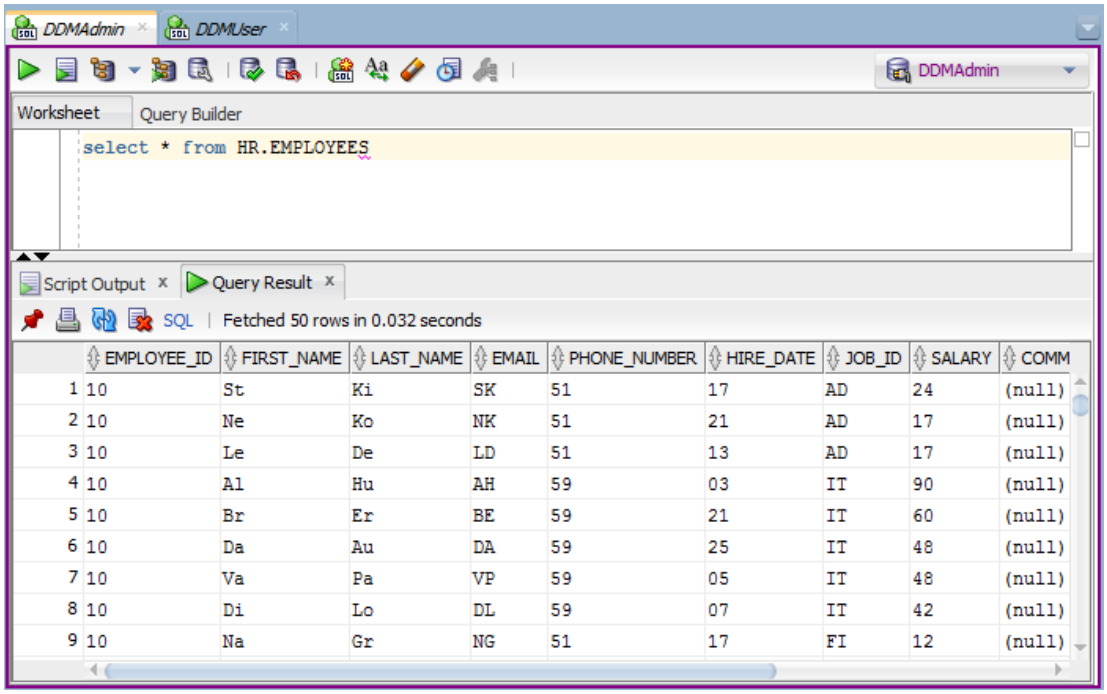
3. Click **Run**.

   The Security Rule Set Simulator displays the altered SQL request in the **Log Output** field.

The following image shows the Security Rule Set Simulator parameters and the altered SQL request in the **Log Output** field:



If the administrator sends the request to the database, Dynamic Data Masking returns masked results. The following image shows the results that the DDMAdmin administrator views from SQL Developer:



Simulating the Rules as a Non-Administrator

Run the Security Rule Set Simulator as a database user to view how the Rule Engine applies the blocking rule to the request.

1. In the Security Rule Set Simulator, leave the parameters from the administrator test except for the following parameter:

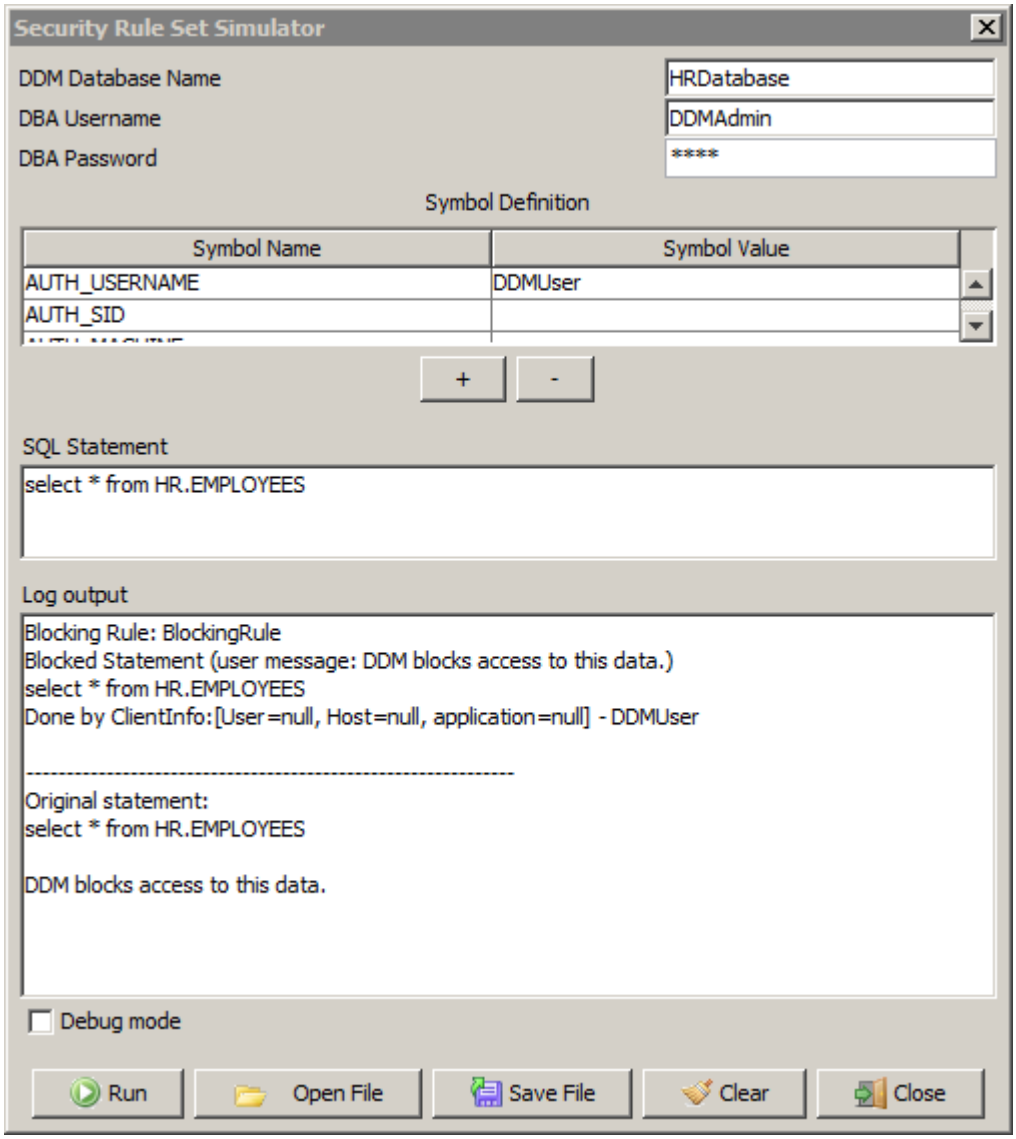
   Symbol Name

   Change the AUTH_USERNAME symbol value to a database user that is not an administrator. In this example, the user is DDMUser.

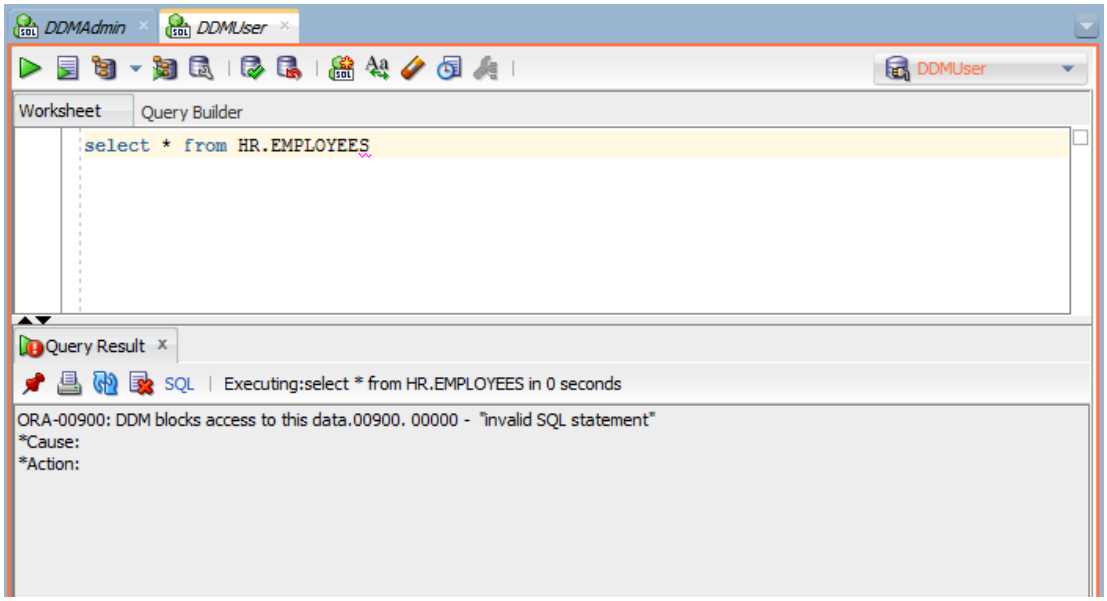
2. In the SQL Statement field, you can enter a new request to test the rule set or use the same request. In this example, the SQL Statement is `select * from HR.EMPLOYEES`.
3. Click **Run**.

   The Rule Engine prevents the user from accessing the sensitive data in the databsae and the Security Rule Set Simulator displays the blocking rule error message in the **Log Output** field. In this example, the error message is `DDM blocks access to this data`.

The following image shows the Security Rule Set Simulator parameters and the log output results:



If the database user sends the request to the database, Dynamic Data Masking returns the blocking rule error message. The following image shows the results that the DDMUser database user views from SQL Developer:



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Acknowledgements

Thanks to the Dynamic Data Masking development and QA teams for their help in completing this article.